Ameya Patil

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I design and build interactive data visualization and analysis systems for human-in-the-loop understanding of **large scale network graphs**, using an interdisciplinary approach of combining learnings from the fields of **Systems, Human Perception and LLMs**.

I am also interested in the topics of **visualization perception research** for both humans and LLMs, and LLM assisted **literacy through data visualizations** for humans, especially in the space of **socio-environmental** data contexts.

Experience

- **University of Washington** — Seattle WA, USA
  PhD Student, advised by **Dr. Leilani Battle** — Sept 2021 - Aug 2026
  Thesis: Evaluation for Scalable and Interactive Network Visualization and Analysis
  - Performed a detailed survey and study of the state of the art in different tools available for managing, visualizing and analyzing network graph data, and synthesized future research directions in network data management and analysis. Work under revision.
  - Designed and implemented a benchmarking framework for evaluating graph systems on interactive network graph visualization and analysis workloads. Detected bugs in popular commercial graph systems using the benchmkarking framework. Work under revision.
  - Performed a design study of domain experts in whale research to design WhaleVis - an interactive analysis dashboard for modeling and analyzing historical whale hunting data as a network graph. Published work in the International Whaling Commission Scientific Committee Meeting 2023 and IEEE InfoVIS 2023.
- **National Center for Atmospheric Research** — Boulder CO, USA
  Data Visualization Intern, advised by **Dr. Helen Kershaw** and **Dr. Moha El Gharamti** — Summer 2023
  - Designed and implemented HydroVis - an interactive analysis dashboard for freshwater flood forecasting using the WRF-Hydro hydrological forecasting model. Published work as a demo in IEEE InfoVIS Viz4Climate + Sustainability Workshop, 2024.
- **AVIZ, Inria** — Saclay, Paris, France
  Research Intern, advised by **Dr. Jean-Daniel Fekete** — Summer 2021
  - Proposed two new bar chart designs for progressive analytics and conducted a crowd-sourced user study to evaluate the new designs and the efficacy of confidence intervals for decision making in progressive analytics. Published work in TVCG 2023.
- **Fraunhofer CESE** — College Park MD, USA
  Research Assistant Intern, advised by **Dr. Marcel Schäfer** — Summer 2019
  - Worked as Java developer on the PocketSecurity project which collects data to analyze user behaviour. Identified and implemented critical data probes to be collected for better analysis, and improved existing probes
- **NVIDIA** — Pune MH, India
  System Software Engineer - C/C++ — July 2016 - July 2018
  - Worked as developer for ShadowPlay - a gameplay sharing app to record, screenshot, broadcast and coplay video games, to develop new multi-threaded and multi-processes features, across software level code and GPU driver code.
  - Enhanced and monitored the automated software testing suite for ShadowPlay and mentored an intern for the same.
- **NVIDIA** — Pune MH, India
  Intern — July 2015 - Dec 2015
  - Device Filter Drivers - C/C++: Implemented end-to-end user input redirection from input devices to a specific application using filter drivers and device notifications
  - Z-buffer - Python: Implemented aesthetic visual effects such as zoom burst using the depth data of images

Skills

- **Programming Languages:** C/C++, Java, Python, Javascript, SQL, Cypher, GSQL
- **Libraries/Frameworks:** D3.js, Matplotlib, Vega-lite, Altair, OpenCV, MPI, OpenMP
- **Soft Skills:** Collaboration, Public Speaking, Mentoring
- **Extra-Curricular:** Digital Photography, Photo and Video Editing

Languages

- **Marathi:** Native
- **Hindi:** Fluent
- **English:** Fluent
- **French:** Intermediate (B2)

Education

- **University of Washington, Seattle** (UW) — WA, USA
  Ph.D. in Computer Science, Advised by **Dr. Leilani Battle**, GPA: 4.00/4.00 — Jan 2021 - Aug 2026
  - **Thesis:** Evaluation for Scalable and Interactive Network Visualization and Analysis
- **University of Maryland, College Park** (UMD) — MD, USA
  M.S. in Computer Science, GPA: 3.84/4.00 — Aug 2018 - Dec 2020
- **Birla Institute of Technology and Science - Pilani** (BITS) — Goa, India
  B.E. (Honors) in Computer Science, GPA: 8.24/10.00 — Aug 2012 - May 2016

Publications

1. **A. Patil**, W. Tan, I. Sinha, L. Battle, "Task-Centered Benchmark for Interactive Network Visualization & Analysis", CoRR, vol. abs/2607.03725, 2026. *arXiv:2607.03725*
2. **A. Patil**, L. Battle, "Scalability ***and*** Interactivity in Network Data Management & Understanding Systems: a Survey", Under revision.
3. Z. Dong, T. Barrett, **A. Patil**, Y.Shoda, L. Battle, E. Wall, "A Design Space of Behavior Change Interventions for Responsible Data Science", Proceedings of the Conference on Intelligent User Interface (IUI), 2025. DOI: 10.1145/3708359.3712140. *pdf*
4. Z. Dong, **A. Patil**, Y.Shoda, L. Battle, E. Wall, "Behavior Matters: An Alternative Perspective on Promoting Responsible Data Science", Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW), 2025. DOI: 10.1145/3710932. *arxiv:2410.17273*
5. **A. Patil**, M. Smith, H. Kershaw, M. El Gharamti, "Interactive Visualization of Ensemble Data Assimilation Forecasts for Freshwater Floods", IEEE VIS Viz4Climate + Sustainability Workshop Demo, 2024. *pdf*
6. **A. Patil**, Z. Rand, T. Branch, L. Battle, "WhaleVis: Visualizing the History of Commercial Whaling", IEEE VIS Short Papers, 2023. DOI: 10.1109/VIS54172.2023.00028. *arxiv:2308.04552*
7. **A. Patil**, Z. Rand, T. Branch, L. Battle, "WhaleVis: A New Visualization Tool for the IWC Catch Database", International Whaling Commission SC/69A/GDR/04, 2023. *archive.iwc.int/SC/69A/GDR/04*
8. **A. Patil**, G. Richer, C. Jermaine, D. Moritz, J.-D. Fekete, "Studying Early Decision Making for Progressive Bar Charts", IEEE Transactions on Visualization and Computer Graphics, 2023. DOI: 10.1109/TVCG.2022.3209426. HAL: *https://hal.science/hal-03738461/*
9. A. Aguinaldo, P.-Y. Chiang, A. Gain, **A. Patil**, K. Pearson and S. Feizi, "Compressing GANs using Knowledge Distillation", CoRR, vol. abs/1902.00159, 2019. *arXiv:1902.00159*